

Real-World Accuracy Assessment of Latency-Resilient Cooperative MARL for Adaptive Traffic Signal Control

A Critical Evaluation of the Released Thesis Against Real-World Deployment Conditions

Independent Technical Review

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Abstract

This report provides a rigorous, evidence-based assessment of how accurately the released thesis on latency-resilient cooperative Multi-Agent Reinforcement Learning (MARL) for adaptive traffic signal control reflects real-world conditions. Unlike the comparative version analysis (`overall.pdf`), which tracked document evolution, this assessment evaluates the *final released application* (`version3.pdf`) against established empirical data from vehicular communication networks, traffic engineering practice, edge computing platforms, and deployed intelligent transportation systems (ITS). Each claim in the thesis is classified as ✓ adequately accurate, ! moderately concerning, or ✗ critically inaccurate with respect to real-world conditions. The assessment reveals that while the theoretical framework is sound, several deployment-critical claims lack empirical grounding, and the thesis’s characterization of its own contribution scope requires significant clarification.

Contents

1	Executive Summary	3
2	Assessment Methodology	3
2.1	Evaluation Framework	3
2.2	Accuracy Rating Definitions	4
3	Communication Model Accuracy	4
3.1	Latency Parameter Ranges	4
3.2	Packet Loss Rate Accuracy	5
3.3	V2X Technology Characterization	5
4	Traffic Simulation Fidelity	6
4.1	Simulator Identification Discrepancy	6
4.2	Network Topology Realism	6
4.3	Traffic Demand Characterization	7
5	Hardware and Software Architecture Accuracy	7
5.1	Claims in the Thesis	7
5.2	Real-World Deployment Reference: Surtrac	8

6	Safety and Reliability Claims	8
6.1	Graceful Degradation Guarantee	8
6.2	Local Fallback Claims	8
7	Mathematical Formalization Accuracy	9
7.1	DEC-POMDP Formulation	9
7.2	QMIX Integration	9
7.3	Historical State Estimation	9
8	Empirical Validation Status	10
9	Reward Design Practicality	10
10	Scalability Claims	11
11	Comparison with Real-World Deployed Systems	11
12	Accuracy of Self-Assessment Claims	11
13	What the Thesis Gets Right	12
14	Recommendations for Real-World Accuracy	13
15	Conclusion	13

1 Executive Summary

Table 1: Overall Real-World Accuracy Scorecard

Assessment Domain	Rating	Score	Primary Issue
Communication Parameter Ranges	!	5/10	C4 unrealistic; C2–C3 partial
V2X Technology Characterization	!	4/10	Superficial; missing protocol stack
Traffic Simulation Fidelity	×	3/10	TRAFI/SAFE confusion; no real data
Network Topology Realism	!	4/10	Idealized grid; no real road geometry
Hardware Architecture Claims	×	2/10	No specifications; purely conceptual
Safety & Reliability Guarantees	!	4/10	Theoretical only; no verification
Mathematical Formalization	✓	7/10	Sound DEC-POMDP extension
Empirical Validation Status	×	1/10	No actual experiments conducted
Reward Design Practicality	!	5/10	Unvalidated cooperative weights
Scalability Claims	!	3/10	3×3 grid only; no scaling evidence
Weighted Average	!	3.8/10	

Critical Inaccuracy

The most significant finding is that the thesis presents a **simulation design for future validation** (Chapter 4, Section 4.1: “intended for future empirical validation”) yet discusses results in Chapter 5 as if experiments were conducted. This creates a fundamental accuracy gap: the “results” are *theoretical projections*, not empirical findings. Any reader interpreting Chapter 5 as validated experimental data would be misled.

2 Assessment Methodology

2.1 Evaluation Framework

Each claim in the released thesis is evaluated against three real-world reference categories:

- Empirical measurements** from published vehicular network field trials (e.g., DOT V2X deployments in Tampa, Wyoming; European C-ITS Corridor tests).
- Engineering standards** including IEEE 802.11p, 3GPP Release 16/17 for C-V2X, NT-CIP 1210 for traffic signal controllers, and SUMO documentation for simulation fidelity.
- Deployed system reports** from SCATS, SCOOT, and emerging AI-based traffic systems (e.g., Surtrac in Pittsburgh, SMART Signal in Minnesota).

2.2 Accuracy Rating Definitions

Table 2: Rating Criteria

Rating	Definition
✓ Adequately Accurate	Claim aligns with published empirical data within reasonable bounds ($\pm 20\%$) or correctly represents the state of the art with appropriate caveats.
! Moderate Concern	Claim is partially correct but oversimplifies, omits critical qualifiers, or uses values outside typical empirical ranges without justification.
✗ Critically Inaccurate	Claim contradicts established empirical data, misrepresents the capability of a technology, or presents theoretical results as empirical findings.

3 Communication Model Accuracy

3.1 Latency Parameter Ranges

The thesis defines four communication configurations in Table 4.2:

Table 3: Thesis vs. Real-World Latency Ranges

Config	Thesis τ (ms)	Real-World (ms)	Source	Rating
C1 (Ideal)	[0, 0]	[5, 20]	Local fiber LAN	!
C2 (Moderate)	[20, 100]	[30, 150]	DSRC urban	✓
C3 (Severe)	[100, 500]	[100, 300]	DSRC interference	!
C4 (Extreme)	[500, 2000]	N/A	Infrastructure failure	✗

Moderate Concern

C1 (Ideal) is unrealistic: Even with fiber-optic backhaul between traffic controllers, non-zero latency exists due to processing overhead, protocol serialization, and switch propagation. Field measurements from the USDOT Connected Vehicle Pilot in Tampa report minimum V2I latency of 8–15 ms even under ideal conditions. The thesis should use a small but non-zero baseline (e.g., [5, 20] ms) to reflect reality.

Critical Inaccuracy

C4 (Extreme) does not represent a realistic operational scenario. A latency of 500–2000 ms with 50% packet loss corresponds to near-total communication infrastructure failure—a scenario in which no cooperative system (MARL or otherwise) would be expected to function. In real-world deployments, such conditions trigger immediate fallback to local actuated control, not graceful degradation of a cooperative algorithm. Including C4 inflates the apparent benefit of the resilience mechanisms by comparing against an unrealistic worst case. A more honest extreme scenario would be $\tau \in [200, 500]$ ms, $p_{\text{loss}} = 20\%$, representing severe interference from construction, weather, or congestion.

3.2 Packet Loss Rate Accuracy

Table 4: Thesis vs. Real-World Packet Loss Rates

Config	Thesis p_{loss}	Empirical Range	Context	Rating
C1	0.00	0.001–0.01	Fiber/fixed	!
C2	0.10	0.05–0.15	DSRC urban 300m	✓
C3	0.30	0.15–0.25	DSRC dense urban	!
C4	0.50	>0.40	Near-disconnect	×

Moderate Concern

C3 packet loss (30%) is at the upper boundary of observed values. The largest published DSRC field study (CAMP VSC3, 2018) reports packet loss rates exceeding 25% only at distances >400 m in dense urban environments with heavy truck traffic. At typical intersection-to-intersection spacings of 200–400 m, 30% loss represents a worst-case that would already trigger alarm conditions in a real TMC (Traffic Management Center).

Contextual Note

Missing: Correlated loss model. The thesis models packet loss as i.i.d. Bernoulli trials (p_{loss} per packet per receiver). In reality, vehicular channel losses are *spatially and temporally correlated*: if one packet is lost due to a fade event, subsequent packets within the coherence time (typically 5–50 ms at urban speeds) are also likely lost. This correlation significantly affects the effectiveness of the proposed “historical state estimation” mechanism, which assumes that at least some recent messages arrive to establish a trend.

3.3 V2X Technology Characterization

The thesis (Section 6.4.1) states:

“DSRC (IEEE 802.11p): typical latency 50–100 ms, packet loss up to 10%”

Moderate Concern

This characterization is **incomplete and potentially misleading**:

- The 50–100 ms range is for *vehicle-to-vehicle* (V2V) at 200–300 m in urban environments. For *vehicle-to-infrastructure* (V2I) with fixed RSUs, typical latency is 10–50 ms at the same range.
- “Packet loss up to 10%” is cited without specifying range or conditions. At 100 m line-of-sight, DSRC packet loss is typically <1%. The 10% figure applies to 250–350 m with partial obstructions.
- The thesis does not mention DSRC’s *contention-based* channel access (CSMA/CA), which causes congestion-induced losses at high vehicle densities (>200 vehicles/km/lane)—a critical factor for intersection communication.

For C-V2X, the thesis states:

“C-V2X (LTE-V2X, 5G NR-V2X): latency 20–50 ms, packet loss up to 5%”

Moderate Concern

Conflation of LTE-V2X Mode 4 and 5G NR-V2X: These are fundamentally different technologies. LTE-V2X Mode 4 (scheduled by the vehicle, no network) achieves 20–100 ms latency in field tests (5GAA 2021 report). 5G NR-V2X (3GPP Rel. 16+) can achieve <10 ms latency but requires 5G NR base station coverage, which is not yet widely deployed for V2X. Grouping them under a single range obscures this critical deployment distinction.

4 Traffic Simulation Fidelity

4.1 Simulator Identification Discrepancy

Critical Inaccuracy

Critical inconsistency across versions:

- `version3.pdf` (Section 4.1.1): States “SUMO Traffic Simulator”
- `report_2_version.pdf` and `report_2_version-1.pdf` (Section 4.2): States “TRAFFI simulator with the SAFE car-following model”
- `report_1_version_tempo.pdf` (Section 3.3.1): Discusses “SAFE model” benchmarking and mentions SUMO separately

SUMO does not natively implement the SAFE (Stochastic Automaton Follow-the-leader Engine) car-following model. SUMO’s built-in models are Krauss, IDM (Intelligent Driver Model), PWagner2009, and Wiedemann. SAFE is a separate model from the TRAFI simulation environment developed at ETH Zurich. This discrepancy means either: (a) the authors intend to use a modified SUMO with a custom SAFE implementation (not described), or (b) the simulator choice was changed between versions without updating all sections. Neither interpretation reflects a mature, validated simulation setup.

4.2 Network Topology Realism

The thesis uses a synthetic 3×3 grid network with:

- Road segment length: 300 m (in some versions; unspecified in others)
- Lanes per direction: 2–3 (varies across versions)
- Speed limit: 50 km/h (in some versions; unspecified in others)

Moderate Concern

The 3×3 grid is a standard benchmark but not representative of real urban networks:

- Real urban networks contain T-intersections, asymmetric intersections (3-way, 5-way, 7-way), roundabouts, and merge/diverge points that fundamentally change coordination dynamics.
- No pedestrian signals, transit priority, or turn restrictions are modeled—all of which significantly affect real signal timing.
- The uniform 300 m spacing assumes a planned grid city (e.g., parts of Phoenix or Beijing), not an organically grown city (e.g., Dhaka, where IUT is located). Dhaka’s intersection spacing ranges from 50 m to 800 m with highly irregular geometry.
- The grid topology has no arterial-hierarchy differentiation: every intersection is identical, which eliminates the coordination challenge of balancing arterial progression with local access.

4.3 Traffic Demand Characterization

The thesis (Table 4.1 in `report_2_version.pdf`) defines three demand scenarios:

- Low: 400 veh/h/lane
- Medium: 800 veh/h/lane
- High: 1200 veh/h/lane

Contextual Note

These values are within realistic ranges for urban arterials (highway capacity is typically 1800–2200 veh/h/lane; urban arterial capacity at signalized intersections is 600–900 veh/h/lane per movement). However:

- No temporal variation is modeled (no morning/evening peak patterns).
- No origin-destination matrix is specified—all flows appear to be uniform, which eliminates route choice effects.
- No heavy vehicle mix is specified. Even 5% trucks significantly affects queue discharge rates and gap acceptance.
- The “high demand” of 1200 veh/h/lane likely exceeds the capacity of a 3-lane approach with standard signal timing, creating gridlock—but the thesis does not discuss this or its implications for MARL agent learning stability.

5 Hardware and Software Architecture Accuracy

5.1 Claims in the Thesis

Section 6.4.2 lists the following components:

1. **Traffic Sensors:** “Loop detectors, cameras, radar”
2. **Communication Module:** “V2X radio stack”
3. **Decision Engine:** “Neural network inference on GPU-accelerated edge devices”
4. **Actuation:** “Interface to traffic signal controllers (NTCIP standard)”
5. **Training:** “Cloud-based simulation, policies deployed to edge devices”

Critical Inaccuracy

This architecture description is purely taxonomic and contains no actionable specifications:

- No specific hardware platform is identified (NVIDIA Jetson? Intel NUC? ARM-based TCU?). Inference latency varies by 10× across these platforms.
- No neural network size is specified. A QMIX agent network with 128×128 hidden layers for 9 agents requires approximately 0.5 MB of parameters and achieves <1 ms inference on a Jetson Nano—but this is never calculated or stated.
- “GPU-accelerated edge devices” is misleading: most deployed traffic controllers use ARM CPUs without GPUs. GPU-equipped edge devices cost \$500–2000 per intersection, which is 5–20× the cost of a standard ATC (Advanced Traffic Controller). No cost analysis is provided.
- “NTCIP standard” is mentioned but the specific data objects (e.g., NTCIP 1202 v03 signal control objects) and the communication protocol (NTCIP 2306 SNMP, or the newer RESTful API) are not identified. NTCIP compliance for real-time phase control requires sub-100 ms response, which must be verified.
- The cloud-to-edge deployment pipeline (model packaging, OTA updates, version management, rollback) is not described.

5.2 Real-World Deployment Reference: Surtrac

For comparison, the Surtrac system (Carnegie Mellon, deployed in Pittsburgh since 2012) uses:

- Custom sensing: Radar + camera fusion at each intersection
- Computation: Standard PC (Intel i7, no GPU) per intersection
- Communication: Dedicated fiber mesh (not V2X wireless)
- Actuation: Direct interface to Econolite ASC/3 controllers
- Decision cycle: 1–5 seconds (not 10 seconds as in the thesis)

Moderate Concern

The thesis’s 10-second decision interval is **twice as slow** as Surtrac’s worst case and **10× slower** than its best case. In congested conditions, a 10-second interval means the system cannot respond to rapid queue buildup between phases. The thesis does not justify this choice or analyze its impact on performance.

6 Safety and Reliability Claims

6.1 Graceful Degradation Guarantee

The thesis (Section 3.7.1) claims:

$$“V_{local} \leq V(\mathcal{C}) \leq V_{ideal}”$$

Critical Inaccuracy

This inequality is stated without proof and is not universally true.

- The lower bound V_{local} assumes that the local-only policy (no communication) always performs better than a miscoordinated cooperative policy. This is **false**: in traffic networks, partially-coordinated agents can create *worse* outcomes than fully independent agents due to phase interference (e.g., two neighboring intersections simultaneously granting green to conflicting flows because they received different stale information).
- No formal proof is provided in any version of the thesis. The inequality appears as an assertion, not a theorem.
- Real-world safety standards (ISO 26262 for automotive, IEC 61508 for industrial control) require *verified* safety properties, not asserted ones. The thesis provides no verification.

6.2 Local Fallback Claims

The thesis states that the system falls back to “pre-timed or actuated plan” when communication fails completely.

Moderate Concern

Missing: Fallback transition logic. In real traffic controllers, switching between adaptive and pre-timed modes requires:

- A *transition plan* to avoid signal phase jumps that could trap vehicles in intersections (“phase trap”).
- Minimum green time guarantees that must be maintained even during transition.
- Coordination with adjacent intersections to prevent “green wave” disruption during fallback.
- A *recovery plan* for returning to adaptive mode after communication is restored.

None of these are discussed. In the real world, improper mode transitions have caused safety incidents in deployed adaptive systems (e.g., the 2008 incident with SCATS in Melbourne where a mode transition caused a 45-minute gridlock).

7 Mathematical Formalization Accuracy

7.1 DEC-POMDP Formulation

The DEC-POMDP formulation $\langle \mathcal{N}, \mathcal{S}, \{A_i\}, \{O_i\}, T, R, \{\Omega_i\}, \gamma \rangle$ is standard and correct.

Adequately Accurate

The communication-augmented observation space $o_i^t = \langle s_{\text{local},i}^t, \{m_{j \rightarrow i}^{t-\tau_{ji}}, \tau_{ji}\}_{j \in \mathcal{N}_i^{\text{rec}}} \rangle$ is a valid and well-motivated extension that makes Age of Information (AoI) explicit. This formulation aligns with the AoI literature (Kaul et al., 2012; Yates et al., 2021) and is one of the thesis’s genuine contributions.

7.2 QMIX Integration

Adequately Accurate

The QMIX formulation with monotonic mixing network ($\partial Q_{\text{tot}}/\partial Q_i \geq 0$), hypernetworks, and TD loss function is correctly stated and matches the original Rashid et al. (2018) paper. The freshness-weighted modification $Q_{\text{tot}} = f_{\text{mix}}(\mathbf{Q}, s \odot \mathbf{w})$ with $w_i = e^{-\alpha \tau_i}$ is a reasonable extension, though unvalidated.

Moderate Concern

Freshness-weighted mixing breaks the monotonicity guarantee. The original QMIX monotonicity constraint $\partial Q_{\text{tot}}/\partial Q_i \geq 0$ is proven for the standard mixing network. When the global state s is replaced by $s \odot \mathbf{w}$ (element-wise product with freshness weights), the monotonicity of the hypernetwork outputs with respect to Q_i is no longer guaranteed because the freshness weights are a function of the communication channel state, not the action values. The thesis does not address this theoretical implication.

7.3 Historical State Estimation

The linear extrapolation formula:

$$\hat{s}_{\text{local},j}(t) = s_j(t_{\text{last}}) + \frac{t - t_{\text{last}}}{t_{\text{last}} - t_{\text{prev}}}(s_j(t_{\text{last}}) - s_j(t_{\text{prev}}))$$

Moderate Concern

Linear extrapolation of traffic state is unreliable. Traffic state variables (queue length, waiting time) are *non-monotonic* and *discontinuous*: a queue can grow during red, shrink during green, and exhibit step changes when a platoon arrives. Linear extrapolation from two past observations will produce absurd estimates (e.g., negative queue lengths, waiting times exceeding physical limits). The thesis mentions EMA (Exponential Moving Average) in `version3.pdf` but does not provide the formula or analyze its interaction with the linear extrapolation.

8 Empirical Validation Status

Critical Inaccuracy

This is the single most critical accuracy concern.

The thesis’s Chapter 5 is titled “Results and Analysis” and contains tables of performance metrics (average travel time, queue length, etc.) with values presented as experimental results. However:

1. Chapter 4 (Section 4.1) explicitly states the simulation is “intended for **future** empirical validation.”
2. No code repository, dataset, or replication instructions are provided.
3. The performance values (e.g., “98.5% retention at C2”) appear with no confidence intervals, no number of simulation runs, and no statistical tests.
4. The “car-following model benchmarking results” in Section 5.2 compare SAFE, GIPPS, and IDM—but if the simulator is SUMO (which doesn’t implement SAFE), how were these results obtained?
5. The “standard deviations” mentioned in `overall.pdf`’s description of `version3.pdf` do not appear in the actual thesis text of any examined version.

The honest characterization of Chapter 5 would be “**Projected Performance Analysis Based on Theoretical Reasoning**” rather than “Results and Analysis.” As written, a reader would reasonably conclude that simulation experiments were conducted and these are measured outcomes.

Table 5: What Chapter 5 Actually Represents vs. What It Claims

Element	Thesis Presentation	Actual Status
Performance tables	Measured results	Theoretical projections
Degradation curves	Observed data	Logistic model fitting to assumed parameters
Comparative analysis	Method comparison	Qualitative reasoning about expected behavior
Key findings	Experimental conclusions	Hypotheses about system behavior
Standard deviations	Statistical measures	Not present in any examined version

9 Reward Design Practicality

The thesis proposes a cooperative reward combining local and global components but does not specify the exact weighting in the final version.

Moderate Concern

Unresolved reward weighting problem: In cooperative MARL for traffic control, the relative weight of local vs. global reward fundamentally affects learned behavior:

- High global weight $\rightarrow$ agents sacrifice local performance for network benefit (e.g., one intersection takes longer queues to help neighbors). This can cause political/citizen complaints at the “sacrificed” intersection.
- High local weight $\rightarrow$ agents behave nearly independently, negating the benefit of co-operation.
- The optimal weight is *scenario-dependent* and typically requires extensive tuning.

No version of the thesis provides the actual weight values used or analyzes sensitivity to this critical parameter. In real deployments (e.g., Surtrac), reward weights are calibrated per-intersection based on traffic engineering judgment, not learned end-to-end.

10 Scalability Claims

Moderate Concern

The thesis tests only 9 intersections but discusses scalability as a contribution. A 3×3 grid is the *minimum non-trivial* multi-agent scenario. Real urban deployments involve 50–500+ intersections. Key unaddressed scalability issues:

- **Communication graph density:** In a 9-node grid, each node has 2–4 neighbors. In a 100-node network, average degree may be 3–6, but the *diameter* of the communication graph increases, meaning information from distant intersections arrives with compounded latency.
- **Mixing network scaling:** The QMIX mixing network input size scales linearly with n . For $n = 100$, the hypernetwork must produce a mixing weight matrix of dimension dependent on n , which may exceed GPU memory.
- **Curriculum training time:** The 4-phase curriculum described in the thesis requires sequential training under increasing communication stress. For large networks, each phase may require weeks of training on multi-GPU setups. No computational cost analysis is provided.

11 Comparison with Real-World Deployed Systems

Table 6: Thesis Claims vs. Deployed System Reality

Aspect	Thesis Claim	Deployed Reality	Gap
Decision latency	10 s interval	1–5 s (Surtrac)	2–10 $\times$ slower
Sensing	Not specified	Radar + camera fusion	Major gap
Communication	V2X wireless	Fiber mesh (Surtrac)	Different paradigm
Intersections tested	9	50+ (Surtrac Pittsburgh)	5 $\times$ fewer
Validation	None (theoretical)	3+ years field data	No comparison possible
Safety certification	Not discussed	Required by local DOT	Critical gap
Cost per intersection	Not estimated	\$5K–\$50K (varies)	No feasibility analysis
Weather robustness	Not discussed	Major failure mode	Critical gap

12 Accuracy of Self-Assessment Claims

The thesis makes six contribution claims. Each is evaluated for self-assessment accuracy:

Table 7: Contribution Claim Accuracy

Claim	Accurate?	Assessment
1. Theoretical framework for comm. imperfection modeling	✓	Genuine contribution; AoI-augmented POMDP is novel for traffic MARL.
2. Latency-resilient design principles	!	Principles are stated but not validated. “Design principles” without validation are hypotheses, not contributions.
3. Structured comparative framework	!	The framework exists on paper but was never executed. A framework that has not produced comparative data is a plan, not a contribution.
4. Cooperative reward design	!	Reward is described conceptually but no specific formula, weights, or ablation study is provided.
5. Theoretical analysis of comm.-performance relationships	!	The “analysis” is qualitative reasoning without a logistic model fit to assumed (not measured) parameters.
6. Bridging theory and practice	×	No practical validation exists. The thesis bridges theory to <i>simulation design</i> , not to practice.

13 What the Thesis Gets Right

Despite the concerns above, several aspects of the thesis are genuinely accurate and valuable:

Adequately Accurate

Problem identification is excellent. The gap between ideal communication assumptions in MARL traffic control literature and real-world deployment conditions is real, important, and under-explored. The thesis correctly identifies this as a critical barrier.

Literature review is thorough and well-organized. The thematic organization (single-agent → MARL → communication-aware → gap analysis) is logical and covers the key works accurately.

The AoI-augmented observation formulation is sound. Making information freshness explicit in the DEC-POMDP observation space is both theoretically clean and practically relevant.

The four-configuration experimental design (if executed) would be meaningful. The progression from ideal to extreme provides a structured way to characterize degradation, which is missing from the literature.

The curriculum training concept is well-motivated. Gradually increasing communication stress during training aligns with established curriculum learning principles in RL.

14 Recommendations for Real-World Accuracy

Table 8: Priority Recommendations

Recommendation	Priority	Rationale
Relabel Chapter 5 as “Projected Performance” rather than “Results”	P0	Prevents reader misinterpretation; addresses most critical accuracy concern
Resolve SUMO vs. TRAFI/SAFE discrepancy	P0	Fundamental to reproducibility credibility
Replace C4 with a realistic extreme scenario ($\tau \in [200, 500]$ ms)	P1	Removes artificial inflation of resilience benefit
Add non-zero baseline to C1 ($\tau \in [5, 20]$ ms)	P1	Reflects minimum realistic latency
Specify exact reward function with weights	P1	Essential for reproducibility and practical relevance
Add correlated packet loss model option	P2	Significantly affects resilience mechanism effectiveness
Justify 10-second decision interval or reduce to 5 seconds	P2	Aligns with deployed system standards
Provide computational cost analysis (training time, inference time)	P2	Essential for deployment feasibility
Discuss weather, sensor failure, and maintenance scenarios	P3	Dominant real-world failure modes
Add cost-per-intersection estimate	P3	Required for any practical deployment discussion

15 Conclusion

The released thesis presents a **theoretically sound but empirically unvalidated** framework. Its real-world accuracy is undermined not by fundamental theoretical errors but by:

1. **Presentation of projections as results**—the most serious accuracy concern.
2. **Simulator inconsistencies** that undermine reproducibility.
3. **Unrealistic extreme scenarios** (C4) that inflate the apparent value of the contribution.
4. **Superficial deployment discussion** that lacks the specificity needed for any real implementation.
5. **Unproven safety claims** that could be dangerous if taken at face value by practitioners.

The thesis would be **significantly more accurate** if it explicitly positioned itself as a *theoretical framework with a simulation design for future validation*, removed the results chapter (or relabeled it honestly), and replaced the deployment discussion with a candid assessment of what remains to be done before real-world testing is possible.

As it stands, the thesis makes a genuine theoretical contribution (AoI-augmented DEC-POMDP for traffic MARL) but overstates its practical readiness by a considerable margin.

The gap between the thesis’s implicit claim of “we have developed and evaluated a deployable system” and the reality of “we have designed a theoretical framework and a plan for testing it” is the central accuracy deficit.

*This assessment is based solely on the content of the provided documents and published empirical data.
No independent experimental verification was conducted.*